Functions & Transformations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 2x + 5$ in transformations of y=x, evaluate $f(5)$.

Final Answer: 15. Substitute $x = 5$: $f(5) = 2(5) + 5 = 15$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 3x + 6$ in slope-intercept form, evaluate $f(2)$.

Final Answer: 12. Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for slope-intercept form consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 20.5. The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 4x + 7$ in standard form, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 4(2) + 7 = 15$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for standard form consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 5x + 8$ in point-slope form, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 5(2) + 8 = 18$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for point-slope form consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Let $f(x) = \frac{2x-7}{5}$. (a) Find $f^{-1}(x)$. (b) Solve the equation $f(f^{-1}(x)) = 9$. (c) Find the value of k for which $f(k) = f^{-1}(k)$.

(a) Find $f^{-1}(x)$: Let $y = \frac{2x-7}{5}$. Swap x and y: $x = \frac{2y-7}{5} \Rightarrow 5x = 2y - 7 \Rightarrow 2y = 5x + 7 \Rightarrow \mathbf{f^{-1}(x) = \frac{5x+7}{2}}$. (b) Composition identity: By definition, for all x in the domain of f^{-1}, $f(f^{-1}(x)) = x$. Therefore, $f(f^{-1}(x)) = 9 \Rightarrow \mathbf{x = 9}$. (c) Point on line $y = x$: If $f(k) = f^{-1}(k)$, the intersection must lie on the line of reflection $y = x$. Setting $f(k) = k$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x-1}{2x+5}$.

Write $y = \frac{3x-1}{2x+5}$. The domain of f is $x \neq -2.5$. Its horizontal asymptote is $y = 3/2$, so the range of f is $y \neq 1.5$. Swap x and y and solve for y: $x = \frac{3y-1}{2y+5} \Rightarrow x(2y+5) = 3y - 1$. $2xy + 5x = 3y - 1 \Rightarrow 2xy - 3y = -5x - 1$. $y(2x-3) = -(5x+1) \Rightarrow y = \frac{-5x-1}{2x-3} = \frac{5x+1}{3-2x}$. Inverse function: $f^{-1}(x) = \frac{5x+1}{3-2x}$. Domain of f^{-1}: $x \neq \frac{3}{2}$ (the range of f). Range of f^{-1}.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

If $\sin\theta = -\frac{4}{5}$ and θ lies in Quadrant III, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Using the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$: $\left(-\frac{4}{5}\right)^2 + \cos^2\theta = 1 \Rightarrow \frac{16}{25} + \cos^2\theta = 1 \Rightarrow \cos^2\theta = \frac{9}{25}$. Since θ is in Quadrant III, cosine is negative: $\cos\theta = -\frac{3}{5}$. Tangent is positive in Quadrant III: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-4/5}{-3/5} = \frac{4}{3}$. Double-angle formula for sine.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find all real and complex roots of the polynomial $P(x) = 2x^3 + 3x^2 - 8x + 3$.

By the Rational Root Theorem, possible rational roots are $\pm\{1, 3\}/\{1, 2\} = \pm\{1, 3, 1/2, 3/2\}$. Test $x = 1$: $P(1) = 2(1)^3 + 3(1)^2 - 8(1) + 3 = 2 + 3 - 8 + 3 = 0$. ✓ So $(x - 1)$ is a factor. Divide $P(x)$ by $(x - 1)$ using synthetic division: $2x^3 + 3x^2 - 8x + 3 = (x - 1)(2x^2 + 5x - 3)$. Factor the remaining quadratic: $2x^2 + 5x - 3 = (2x - 1)(x + 3)$. Setting each factor to zero yields the complete set of roots: $x = 1, \quad x = \frac{1}{2}, \quad x = -3$.

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12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Determine the maximal real domain of the function $f(x) = \frac{\sqrt{3x-12}}{x^2-25}$. Express your answer in interval notation.

Step 1: For the square root in the numerator to be defined over the reals, the radicand must be non-negative: $3x - 12 \geq 0 \Rightarrow 3x \geq 12 \Rightarrow x \geq 4$. Step 2: For the denominator to be non-zero: $x^2 - 25 eq 0 \Rightarrow x eq 5 \quad \text{and} \quad x eq -5$. Step 3: Intersect the conditions $x \geq 4$ and $x eq 5$ (note that $x = -5$ is already outside $x \geq 4$): $[4, 5) \cup (5, \infty)$. The maximal real domain is **$[4, 5) \cup (5, \infty)$**.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In triangle OAB, $\vec{OA} = 3\mathbf{a}$ and $\vec{OB} = 6\mathbf{b}$. Point P lies on AB such that $AP : PB = 1 : 2$. Find $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ in its simplest form.

Step 1: Express $\vec{AB}$: $\vec{AB} = \vec{AO} + \vec{OB} = -3\mathbf{a} + 6\mathbf{b}$ Step 2: Find $\vec{AP}$: $\vec{AP} = \frac{1}{3}\vec{AB} = \frac{1}{3}(-3\mathbf{a} + 6\mathbf{b}) = -\mathbf{a} + 2\mathbf{b}$ Step 3: Find $\vec{OP}$: $\vec{OP} = \vec{OA} + \vec{AP} = 3\mathbf{a} + (-\mathbf{a} + 2\mathbf{b}) = 2\mathbf{a} + 2\mathbf{b}$

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The graph of $y = f(x)$ passes through the points $(-2, 8)$ and $(4, -6)$. If $g(x) = 3f(x - 5) + 2$, determine the corresponding points on the graph of $g(x)$.

Transformation Mapping: For any point (x, y) on f, the corresponding point on $g(x) = 3f(x - 5) + 2$ has coordinates: $(x', y') = (x + 5, 3y + 2)$. For point 1: $(-2, 8)$:

$x'_1 = -2 + 5 = 3, \quad y'_1 = 3(8) + 2 = 24 + 2 = 26 \Rightarrow$ **$(3, 26)$**. For point 2: $(4, -6)$:

$x'_2 = 4 + 5 = 9, \quad y'_2 = 3(-6) + 2 = -18 + 2 = -16 \Rightarrow$ **$(9, -16)$**. The corresponding points on $g(x)$ are **$(3, 26)$** and **$(9, -16)$**.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 9.01 Radicals & Exponents**

Given $f(x) = x^2 - 4x + 7$ with restricted domain $x \geq 2$, find an explicit formula for $f^{-1}(x)$ and state its domain.

Step 1: Complete the square to isolate x: $y = (x^2 - 4x + 4) + 3 = (x - 2)^2 + 3$. Step 2: Swap x and y (with $y \geq 2$): $x = (y - 2)^2 + 3 \Rightarrow (y - 2)^2 = x - 3$. Step 3: Take the principal square root (since $y \geq 2 \Rightarrow y - 2 \geq 0$): $y - 2 = +\sqrt{x - 3} \Rightarrow y = 2 + \sqrt{x - 3}$. Therefore, **$\mathbf{f^{-1}(x) = 2 + \sqrt{x - 3}}$**, defined for domain **$\mathbf{x \geq 3}$** (which matches the range of f).

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Linear Equations & Lines**

Solve the exponential equation $4^{x+1} - 5(2^x) + 1 = 0$ for all real values of x.

Rewrite the first term in terms of base 2: $4^{x+1} = 4^1 \cdot (2^2)^x = 4 \cdot (2^x)^2$. Substitute $u = 2^x$ (with constraint $u > 0$): $4u^2 - 5u + 1 = 0$. Factor the quadratic: $(4u - 1)(u - 1) = 0 \Rightarrow u = \frac{1}{4}$ or $u = 1$. Back-substitute $2^x = u$:$2^x = \frac{1}{4} = 2^{-2} \Rightarrow x = -22^x = 1 = 2^0 \Rightarrow x = 0$Both solutions are real and valid: $x = -2$ and $x = 0$.

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Rational Expressions**

Section I of EST I Math contains 20 questions in 25 minutes. If a student answers the first 12 questions in an average of 50 seconds each, how many seconds per question are available for the remaining 8 questions?

Time for first 12 questions = 12 × 50 = 600 seconds (10 minutes). Total time = 25 × 60 = 1500 seconds. Remaining time = 1500 − 600 = 900 seconds. For the remaining 8 questions: $\frac{900}{8} = 112.5$ seconds per question (1 minute 52.5 seconds per question).

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Statistics & Data Analysis**

A student has scores of 78, 85, 92, and 81 on four exams. What score must the student earn on the fifth exam to achieve an overall arithmetic mean of 85 across all five exams?

Step 1: Determine the required sum of all 5 exam scores: $\sum x = 5 \times 85 = 425$. Step 2: Sum the first four exam scores: $78 + 85 + 92 + 81 = 336$. Step 3: Subtract the accumulated total from the required total to find the 5th score x_5: $x_5 = 425 - 336 = \mathbf{89}$. The student must score 89 on the fifth exam.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratic Functions**

Level 4 · SynthesisLine ℓ passes through $(2, -1)$ and is perpendicular to the line $3x + 6y = 18$. If line ℓ also passes through $(k, 9)$, what is the value of k?

Rewrite $3x + 6y = 18$ in slope-intercept form: $6y = -3x + 18 \Rightarrow y = -\frac{1}{2}x + 3$, so its slope is $-\frac{1}{2}$. The perpendicular line ℓ has slope $m_\ell = 2$. Using $(2, -1)$ and $(k, 9)$: $\frac{9-(-1)}{k-2} = 2 \Rightarrow \frac{10}{k-2} = 2 \Rightarrow k - 2 = 5 \Rightarrow k = 7$.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Exponents & Polynomials

Level 1 · FoundationsThe Digital SAT Math section allocates 70 minutes across two modules of equal length. If each module contains 22 questions, what is the exact average time available per question in seconds?

Step 1: Calculate total seconds per module: $35 \text{ minutes} \times 60 \text{ s/min} = 2100 \text{ seconds}$. Step 2: Divide by the 22 questions in the module: $\frac{2100 \text{ seconds}}{22 \text{ questions}} = \frac{1050}{11} \approx 95.45$ seconds (about 1 min 35 s per question).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 3$ in review: slope as rate of change, evaluate $f(2)$.

a. 13 ✓

b. 18

c. 10

d. 17

Substitute $x = 2$: $f(2) = 5(2) + 3 = 13$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 4$ in the slope formula, evaluate $f(2)$.

a. 16 ✓

b. 22

c. 12

d. 20

Substitute $x = 2$: $f(2) = 6(2) + 4 = 16$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: slope as rate of change consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for the slope formula consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An arithmetic sequence has first term -9 and common difference -3. Find the 10th term.

a. -33

b. -37

c. **-36** ✓

d. -39

$a_{10} = a_1 + (n-1)d = -9 + (10-1)(-3) = -36$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Graph the solution of $x + y \le 4$ and $x - y \le 2$. Which point is in the solution region?

a. $(0, 5)$

b. $(3, 2)$

c. $(1, 1)$ ✓

d. $(5, 0)$

$(1, 1)$: $1 + 1 = 2 \le 4$ ✓ and $1 - 1 = 0 \le 2$ ✓.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A student needs at least an 85 average across 4 exams. If the first three scores are 82, 88, and 79, what is the minimum score required on the fourth exam?

a. 89

b. **91** ✓

c. 93

d. 95

Total points needed: $4 \times 85 = 340$. Sum of first three: $82 + 88 + 79 = 249$. Minimum fourth score: $340 - 249 = 91$.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A researcher models exam scores with the line of best fit $y = 5.2x + 58$, where x is the number of hours studied and y is the predicted score. A student who studied 5 hours earned a score of 88. What is the residual (actual score minus predicted score) for this student?

a. **+4 ✓**

b. -4

c. +6

d. +8

Predicted score: $y = 5.2(5) + 58 = 26 + 58 = 84$. Residual $=$ actual $-$ predicted $= 88 - 84 = +4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

State and apply the intermediate value theorem: if $f(1) = -3$ and $f(4) = 5$, continuous on $[1, 4]$, then:

a. No conclusion possible

b. **There exists $c \in (1, 4)$ with $f(c) = 0$ ✓**

c. $f(2) = 1$

d. f is constant

IVT: continuous function on $[a, b]$ attains every value between $f(a)$ and $f(b)$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $y = -2f(x + 1) - 3$ applies which transformations to $y = f(x)$?

a. Left 1, stretch 2, down 3 (no reflect)

b. Right 1, reflect, stretch 2, up 3

c. Right 1, stretch 2, down 3

d. **Left 1, reflect over x-axis, vertical stretch 2, down 3 ✓**

Inside: $x + 1 \rightarrow$left 1. Outside: $-2 \cdot \rightarrow$reflect+stretch×2, $-3 \rightarrow$down 3.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The Intermediate Value Theorem states that if $P(a)$ and $P(b)$ have opposite signs, then:

a. There are two roots

b. **There is a root between a and b** ✓

c. $P(0) = 0$

d. $P(a)P(b) = 0$

IVT guarantees at least one root in (a, b).

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

On a multiple-choice question with 4 answer choices, what is the probability of guessing the correct answer at random?

a. 0.20

b. **0.25** ✓

c. 0.33

d. 0.50

1 correct out of 4: $1/4 = 0.25$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A function is defined by $f(x) = x^2 - 6x + 5$ for $x \geq 3$. \\ The graph of $y = f(x)$ is a parabola. A straight line with equation $y = k$ intersects the graph of $y = f(x)$ at exactly one point. \\ Find the possible value(s) of k.

a. $k = -4$

b. $k = 5$

c. $k = -4$ or $k = 5$ ✓

d. $k \geq 5$

The function is given by $f(x) = x^2 - 6x + 5$ for $x \geq 3$. \\ To find the vertex of the parabola, we can complete the square: $f(x) = (x^2 - 6x + 9) - 9 + 5 = (x - 3)^2 - 4$. \\ The vertex of the parabola is at $(3, -4)$. \\ Since the domain is restricted to $x \geq 3$, the graph of $y = f(x)$ starts at its vertex and extends upwards as x increases. \\ A horizontal line $y = k$ intersects the graph of $y = f(x)$ at exactly one point if: \\ 1. The line passes through the vertex. In this case, $k = -4$. \\ 2. The.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?

a. 24 ✓

b. 18

c. 17

d. 14

We solve the problem in two key phases: identifying the polynomial $f(x)$ from its graphical features, and applying the Polynomial Remainder Theorem to determine $R(x)$. **Step 1: Determine $f(x)$** - Because the graph is tangent to the x-axis at $x = -2$, $(x + 2)$ is a factor with an even multiplicity of at least 2. - Because the graph crosses the x-axis at $x = 1$ and $x = 3$, $(x - 1)$ and $(x - 3)$ are factors with odd multiplicities of at least 1. - Since $f(x)$ is of degree 4, the sum of multipliciti.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of a polynomial function $f(x)$ is shown below. Which of the following could be the equation for $f(x)$?

a. $f(x) = (x + 2)(x - 1)(x - 3)$ ✓

b. $f(x) = (x - 2)(x + 1)(x + 3)$

c. $f(x) = -(x + 2)(x - 1)(x - 3)$

d. $f(x) = (x + 2)(x - 1)^2(x - 3)$

To determine which polynomial function matches the graph, we analyze the intercepts, multiplicities, and end behavior: 1. **Identify the x-intercepts (zeros):** The graph crosses the x-axis at $x = -2$, $x = 1$, and $x = 3$. By the Factor Theorem, if c is a zero of $f(x)$, then $(x - c)$ is a factor. Therefore, the corresponding linear factors are: $(x - (-2)) = (x + 2)$, $(x - 1)$, and $(x - 3)$ 2. **Multiplicity:** At each of the three x -intercepts, the graph passes directly through.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?

a. $-\frac{5}{2}$ ✓

b. $-\frac{7}{2}$

c. $-\frac{13}{4}$

d. $\frac{3}{2}$

To determine $f(-1)$, we first find the constants a, b, and c: 1. **Horizontal Asymptote:** For an exponential function of the form $f(x) = a \cdot b^x + c$, as the exponential term approaches zero, the function approaches its horizontal asymptote $y = c$. Given that the asymptote is $y = -4$, we have: $c = -4$ 2. **Use the Point $(0, -1)$:** Substitute $x = 0$ and $f(0) = -1$: $f(0) = a \cdot b^0 - 4 = -1$ $a(1) - 4 = -1 \Rightarrow a = 3$ 3. **Use the Point $(2, 8)$:** Substitute $x = 2$ and $f(2) = 8$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the function $f(x) = -(x - 3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$?

a. -5

b. 3

c. 4 ✓

d. 7

The maximum value of a function refers to the greatest output value (or y-value) that the function attains across its domain. The given quadratic function is written in vertex form, $f(x) = a(x - h)^2 + k$, where $a = -1$, $h = 3$, and $k = 4$. 1. For any real number x, the squared term is non-negative: $(x - 3)^2 \geq 0$. 2. Multiplying by -1 reverses the inequality: $-(x - 3)^2 \leq 0$. 3. Adding 4 to both sides gives: $-(x - 3)^2 + 4 \leq 4$, which means $f(x) \leq 4$ for all x. When $x = 3$, the funct.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Statistics & Data Analysis

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?

a. -4

b. -2 ✓

c. -1

d. 1

We solve the problem in three steps: 1. **Determine the quadratic function $f(x)$:** The vertex is given as $(h, k) = (2, 9)$, so the function has the form: $f(x) = a(x - 2)^2 + 9$ Since the graph passes through the point $(0, 5)$: $5 = a(0 - 2)^2 + 9 \Rightarrow 4a + 9 = 5 \Rightarrow 4a = -4 \Rightarrow a = -1$ Expanding $f(x)$:
$f(x) = -(x - 2)^2 + 9 = -(x^2 - 4x + 4) + 9 = -x^2 + 4x + 5$ 2. **Determine the linear function $g(x)$:** The line $g(x) = mx + 1$ passes through the vertex $(2, 9)$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratic Functions

The graph of the function $y = ax^2 + bx + c$ is shown below. The function passes through the points $(1, 0)$ and $(5, 0)$, and its vertex has a y-coordinate of -4. Determine the value of b.

a. -6

b. **6** ✓

c. -4

d. 4

The roots of the quadratic function are given as $x = 1$ and $x = 5$. Therefore, the quadratic can be written in the form $y = a(x - 1)(x - 5)$. The vertex of a parabola is located at the midpoint of its roots, so the x-coordinate of the vertex is $x_v = \frac{1 + 5}{2} = 3$. We are given that the y-coordinate of the vertex is -4. Substituting the vertex coordinates into the factored form of the quadratic: $-4 = a(3 - 1)(3 - 5)$
$-4 = a(2)(-2)$ $-4 = -4a$ $a = 1$ Now, substitute $a = 1$ back into the factored form.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Exponents & Polynomials

The graph of the quadratic function $g(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(-2, 10)$ and $(4, 10)$, and its vertex is at $(1, -8)$. Determine the value of a.

a. 1

b. **2** ✓

c. -1

d. -2

The vertex of a parabola $g(x) = ax^2 + bx + c$ is given by $(h, k) = (-\frac{b}{2a}, g(-\frac{b}{2a}))$. From the graph, the vertex is $(1, -8)$, so $h = 1$ and $k = -8$. Since $h = 1$, we have $-\frac{b}{2a} = 1$, which implies $b = -2a$. The function can be written in vertex form as $g(x) = a(x - h)^2 + k$. Substituting the vertex coordinates $(1, -8)$, we get: $g(x) = a(x - 1)^2 - 8$. The parabola passes through the point $(-2, 10)$. Substitute these coordinates into the vertex form equation: $10 = a(-2 - 1)^2 - 8$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The diagram shows the graph of the quadratic function $f(x) = x^2 + ax + b$. The graph passes through the points $(-1, 8)$ and $(3, 0)$. Find the values of a and b.

a. $a = -2, b = 3$ ✓

b. $a = 2, b = -3$

c. $a = -4, b = 1$

d. $a = 4, b = -1$

The function is given by $f(x) = x^2 + ax + b$. Since the graph passes through $(-1, 8)$, we have $f(-1) = 8$. Substituting $x = -1$ into the function: $(-1)^2 + a(-1) + b = 8$ $1 - a + b = 8$ $-a + b = 7$ (*) Since the graph passes through $(3, 0)$, we have $f(3) = 0$. Substituting $x = 3$ into the function: $(3)^2 + a(3) + b = 0$ $9 + 3a + b = 0$ $3a + b = -9$ (**) Now we have a system of two linear equations with two variables: 1) $-a + b = 7$ 2) $3a + b = -9$ Subtract equation (1) from equation (2).

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Systems & Inequalities

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?

a. -10

b. -6

c. 6

d. **10** ✓

Step 1: Write the equation of $f(x)$ in vertex form. Since the vertex is $(h, k_v) = (1, 8)$, the quadratic function has the form: $f(x) = a(x - 1)^2 + 8$ Step 2: Use the known point $(3, 0)$ to solve for a.
$f(3) = a(3 - 1)^2 + 8 = 0 \Rightarrow 4a + 8 = 0 \Rightarrow a = -2$ Thus, $f(x) = -2(x - 1)^2 + 8$. Step 3: Apply the condition that $g(x)$ passes through $(0, 0)$. $g(0) = f(0 - 2) + k = 0 \Rightarrow f(-2) + k = 0$ Step 4: Evaluate $f(-2)$.
$f(-2) = -2(-2 - 1)^2 + 8 = -2(-3)^2 + 8 = -2(9) + 8 = -10$ Step.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Sequences & Series

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(1, 0)$ and $(5, 0)$. If the vertex of the parabola is at $(3, -4)$, what is the value of a?

a. 1

b. **-1** ✓

c. 2

d. -2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. Since the parabola has x-intercepts at $(1, 0)$ and $(5, 0)$, we can write the function in factored form: $f(x) = a(x - 1)(x - 5)$. The x-coordinate of the vertex of a parabola in this form is the midpoint of the x-intercepts, which is $\frac{1+5}{2} = 3$. This matches the given x-coordinate of the vertex. We are also given that the vertex is at $(3, -4)$. We can substitute these coordinates into the factored form of the function to find the value.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Factoring Polynomials

The graph of the linear function $y = mx + b$ is shown below. If the graph passes through the point $(3, 7)$ and the slope m is $\frac{1}{2}$, what is the value of b?

a. $\frac{1}{2}$

b. $\frac{11}{2}$

c. $\frac{13}{2}$ ✓

d. $-\frac{1}{2}$

The equation of a line is given by $y = mx + b$, where m is the slope and b is the y-intercept. We are given that the line passes through the point $(3, 7)$ and the slope $m = \frac{1}{2}$. We can substitute these values into the equation to solve for b: $7 = (\frac{1}{2})(3) + b$ Now, we simplify and solve for b: $7 = \frac{3}{2} + b$ Subtract $\frac{3}{2}$ from both sides: $b = 7 - \frac{3}{2}$ To subtract, we find a common denominator: $b = \frac{14}{2} - \frac{3}{2}$ $b = \frac{14-3}{2}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Radicals & Exponents

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(-1, 0)$ and $(3, 0)$. The vertex of the parabola is at $(1, -4)$. Determine the value of a.

a. -1

b. 1 ✓

c. 2

d. -2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. Since the parabola has roots at $x = -1$ and $x = 3$, we can write the function in factored form as $f(x) = a(x - (-1))(x - 3) = a(x + 1)(x - 3)$. The vertex of the parabola is given as $(1, -4)$. This means that when $x = 1$, $f(x) = -4$. Substitute the vertex coordinates into the factored form of the function: $-4 = a(1 + 1)(1 - 3)$ $-4 = a(2)(-2)$ $-4 = -4a$ Divide both sides by -4 to solve for a: $a = \frac{-4}{-4}$ $a = 1$ Therefore, the v.

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